

PAPER_1615_NUCLEAR_PB208_BE_A_7_8675

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PAPER_1615 — Pb-208 Binding Energy per A = Pb-208 $BE/A = F \cdot K + \beta + \beta^2 - F \cdot \beta^3 + 3 \approx 7.869 \text{ MeV}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Nuclear

Date: June 18, 2026

Status: CLOSED — 0.020% residual

Observation

PAPER_1209II_S670 (Nuclear Unified Proof Set) gives the closed-form expression:

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$$\text{Pb-208 } BE/A = F \cdot K + \beta + \beta^2 - F \cdot \beta^3 + 3 \approx 7.869 \text{ MeV}$$

``

Residual to AME2020: **0.020%**.

UQFF Closed Identity

``

$$\text{Pb-208 } BE/A = F \cdot K + \beta + \beta^2 - F \cdot \beta^3 + 3 \approx 7.869 \text{ MeV } 0.020\%$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental nuclear measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209II_S670

- Related: PAPER_1494-1605 (other session-2026-06-18 closures)

- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_pb208_be_a_7_8675"})`

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